Digital Filters

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Contents

Digital Filters and Linear Filters

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  Input, x_n —— $\boxed{\text{Digital Filter}}$ —— Output, y_n

- The most commonly used filters are Linear Filter. These have a linear relationship (straight line through the origin) between input and output **sequences**.

Advantages

Digital filters offer many gains over analogue filters, some are listed below.

1. Very difficult to realise analogue filters with low cut-off frequencies.
2. Easy to change characteristics of filter.
3. No ageing/degradation or temperature dependencies.
4. Can be VERY sophisticated, and not too expensive.

Disadvantages

1. Require fast digital processors and processing.
2. The processors require power in order to complete their computation. (Digital Filters have no passive components)

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This is Digital Filtering. The filter is described using the following notation : the current number is x_n, and the previous one is x_{n-1}. If a number x_{n-1} is subtracted from an adjacent number x_n;mathematically the output y_n can be written as

$$y_n = x_n - x_{n-1}$$

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Digital Flters are best analysed using the $\mathbb{Z}$ Transform.

$\mathbb{Z}$ Transform

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$$\mathbb{Z}\{x(n)\} = \sum_{n=0}^{\infty} x(n) Z^{-n}$$

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  where Z is a complex number.
- The transformation is simply done by replacing x_{n-i} with $X \cdot Z^{-i}$ where i is any integer and X is the representation of x in the $\mathbb{Z}$ Domain. Remember $Z^0 = 1$.

Example 1 (again)

Starting with

$$y_n = x_n - x_{n-1}$$

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Starting with

$$y_n = x_n - x_{n-1}$$

This equation can be transformed into the $\mathbb{Z}$ Domain to give

$$\begin{aligned} Y \cdot Z^0 &= X \cdot Z^0 - X \cdot Z^{-1} \\ Y &= X - X \cdot Z^{-1} \\ Y &= X(1 - Z^{-1}) \end{aligned}$$

This results in a description between the input X and the output Y in the $\mathbb{Z}$ Domain.

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- Z is just a complex number; and the description between input and output evaluates to a complex number.
- Of interest are all possible values of Z that lie on the circle of radius 1.

These points are illustrated in the following examples.

Example 2

Suppose a digital filter performs

$$y_n = -x_n + 2x_{n-1} - x_{n-2}$$

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The ratio of $\frac{Y}{X}$ is then

$$-1 + 2Z^{-1} - Z^{-2}$$

Example 3

Suppose a digital filter performs

$$y_n = x_n + 2x_{n-1} + 3x_{n-2} + 2x_{n-3} + x_{n-4}$$

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The ratio of $\frac{Y}{X}$ is then

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Impulse Response

A Digital Impulse is a sequence that contains a single unity element followed by zeros. The impulse response is the response of a filter (the output from the filter) if the input was an impulse.

Input, $x_n = [1, 0, 0, 0, 0, 0, \cdots]$ — $\boxed{\text{Digital Filter}}$ — Output, y_n

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Input, $x_n = [1, 0, 0, 0, 0, 0, \cdots]$ —— Digital Filter —— Output, y_n

- If the impulse response of a filter results in an output that has a finite number of non-zero outputs the filter is called a Finite Impulse Response (FIR) filter.

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- If the impulse response of a filter results in an output that has a finite number of non-zero outputs the filter is called a Finite Impulse Response (FIR) filter. Thus if the sequence y_n in the figure above is a finite length sequence; the filter is a FIR filter
- Filters that create an infinite number of non-zero outputs and are called Infinite Impulse Response Filters (IIR Filters).

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Transfer Function

The $\mathbb{Z}$ Domain relationship between input and output sequences is the transfer function of the filter.

- This results in a complex number describing the relationship between input and output.
- This complex number is also best viewed as having magnitude and phase.
- The unit circle in the $\mathbb{Z}$ Domain is the complex number $1e^{j\theta}$
- θ on the unit circle is the Fourier Transform and is equal to $2\pi\frac{f}{F_s}$, where f is the frequency and F_s is the sampling frequency.

$$Z = e^{j2\pi\frac{f}{F_s}} = e^{j\theta}$$

Also $-\pi \leq \theta \leq \pi$ or $0 \leq \theta \leq 2\pi$.

Magnitude Response On Unit Circle

- This describes the relationship between the magnitude of the output to the magnitude of the input.
- Filters can amplify, attenuate or leave unchanged the magnitude of the input depending on the value of Z.

Phase Response On Unit Circle

- The Phase Response represents the phase offset between the input and output from the filter.
- Filters that operate with real numbers must have phase responses that are anti–symmetric about $\theta = 0$.

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- The Phase Response represents the phase offset between the input and output from the filter.
- Filters that operate with real numbers must have phase responses that are anti–symmetric about $\theta = 0$. This means for a given sampling frequency F_s, the phase response at a frequency f must be equal to the negative value of the phase response at a frequency $-f$.

Pole and Zero

- A Pole in the $\mathbb{Z}$ Domain is a value of Z for which the magnitude response is infinite.
- A Zero in the $\mathbb{Z}$ Domain is a value of Z for which the magnitude response is zero.

The number of Pole and Zero's are always the same.

Example 4

Suppose a digital filter is specified with the following transfer function

$$\frac{Y}{X} = \frac{(Z-(0.5+j0.5))(Z-(0.5-j0.5))(Z-(-1))}{(Z-(j0.9))(Z-(-j0.9))(Z-(-0.9))}$$

We can evaluate the manner of combining x_n and y_n as follows

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\begin{aligned}
\frac{Y}{X} &= [Z^2 - (0.5 - j0.5)Z - (0.5 + j0.5)Z + \\
&\quad (0.5 + j0.5)(0.5 - j0.5)] \times \\
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&\quad \times \frac{[(Z - (-1))]}{(Z - (j0.9))(Z - (-j0.9))(Z - (-0.9))} \\
&= [Z^2 - 0.5Z + j0.5Z - 0.5Z - j0.5Z + 0.5^2 + 0.5^2] \times \\
&\quad \times \frac{(Z + 1)}{(Z - (j0.9))(Z - (-j0.9))(Z - (-0.9))}
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&= [Z^2-0.5Z+j0.5Z-0.5Z-j0.5Z+0.5^2+0.5^2]\times\\
&\qquad \times\frac{(Z+1)}{(Z-(j0.9))(Z-(-j0.9))(Z-(-0.9))}\\
&= \frac{[Z^2-Z+0.5][(Z+1)]}{(Z-(j0.9))(Z-(-j0.9))(Z-(-0.9))}
\end{aligned}$$

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$$\frac{Y}{X} = \frac{Z^3 - Z^2 + 0.5Z + Z^2 - Z + 0.5}{(Z - (j0.9))(Z - (-j0.9))(Z - (-0.9))}$$

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&= \frac{Z^3 - 0.5Z + 0.5}{(Z^2 + (j0.9)Z - (j0.9)Z + 0.9^2)(Z-(-0.9))}
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&= \frac{Z^3 - 0.5Z + 0.5}{(Z-(j0.9))(Z-(-j0.9))(Z-(-0.9))} \\
&= \frac{Z^3 - 0.5Z + 0.5}{(Z^2+(j0.9)Z-(j0.9)Z+0.9^2)(Z-(-0.9))} \\
&= \frac{Z^3 - 0.5Z + 0.5}{(Z^2+0.81)(Z+0.9)}
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&= \frac{Z^3 - 0.5Z + 0.5}{(Z^2 + (j0.9)Z - (j0.9)Z + 0.9^2)(Z - (-0.9))} \\
&= \frac{Z^3 - 0.5Z + 0.5}{(Z^2 + 0.81)(Z + 0.9)} \\
&= \frac{Z^3 - 0.5Z + 0.5}{Z^3 + 0.9Z^2 + 0.81Z + 0.729}
\end{aligned}
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$$\frac{Y}{X} = \frac{Z^3 - 0.5Z + 0.5}{Z^3 + 0.9Z^2 + 0.81Z + 0.729}$$

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$$= \frac{(Z^3 - 0.5Z + 0.5)/Z^3}{(Z^3 + 0.9Z^2 + 0.81Z + 0.729)/Z^3}$$

- Rewriting this using only negative powers of Z,

$$\begin{aligned}
\frac{Y}{X} &= \frac{Z^3 - 0.5Z + 0.5}{Z^3 + 0.9Z^2 + 0.81Z + 0.729} \\
&= \frac{(Z^3 - 0.5Z + 0.5)/Z^3}{(Z^3 + 0.9Z^2 + 0.81Z + 0.729)/Z^3} \\
&= \frac{1 - 0.5Z^{-2} + 0.5Z^{-3}}{1 + 0.9Z^{-1} + 0.81Z^{-2} + 0.729Z^{-3}}
\end{aligned}$$

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$$Y(1 + 0.9Z^{-1} + 0.81Z^{-2} + 0.729Z^{-3}) = \\ X(1 - 0.5Z^{-2} + 0.5Z^{-3})$$

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X - 0.5XZ^{-2} + 0.5XZ^{-3}
\end{aligned}
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$$Y(1 + 0.9Z^{-1} + 0.81Z^{-2} + 0.729Z^{-3}) = X(1 - 0.5Z^{-2} + 0.5Z^{-3})$$

$$Y + 0.9YZ^{-1} + 0.81YZ^{-2} + 0.729YZ^{-3} = X - 0.5XZ^{-2} + 0.5XZ^{-3}$$

$$Y = X - 0.5XZ^{-2} + 0.5XZ^{-3} - 0.9YZ^{-1} - 0.81YZ^{-2} - 0.729YZ^{-3}$$

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X - 0.5XZ^{-2} + 0.5XZ^{-3}
\end{aligned}
$$

$$Y = X - 0.5XZ^{-2} + 0.5XZ^{-3} - 0.9YZ^{-1} - 0.81YZ^{-2} - 0.729YZ^{-3}$$

$$y_n = x_n - 0.5x_{n-2} + 0.5x_{n-3} - 0.9y_{n-1} - 0.81y_{n-2} - 0.729y_{n-3}$$

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Stability in the $\mathbb{Z}$ Domain

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- If this criteria is not met, the the implication is that the summation in the definition of the $\mathbb{Z}$ transform will become infinite.

$$\mathbb{Z}\{x(n)\} = \sum_{n=0}^{\infty} x(n)Z^{-n}$$

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- If this criteria is not met, the the implication is that the summation in the definition of the $\mathbb{Z}$ transform will become infinite.

$$\mathbb{Z}\{x(n)\} = \sum_{n=0}^{\infty} x(n) Z^{-n}$$

Note the stability of a filter cannot be predicted by looking at the magnitude and phase response of the filter on the unit circle. Only the location of the poles in the entire $\mathbb{Z}$ Domain determine the stability of the filter.

Example 5

Suppose a digital filter is specified with the following transfer function

$$\frac{(Z-(0.5+j0.5))(Z-(0.5-j0.5))(Z-(-1))}{(Z-(j0.9))(Z-(-j0.9))(Z-(-1.1))}$$

Determine the relationship between y_n and x_n. Note this filter is unstable;

Example 5

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Determine the relationship between y_n and x_n. Note this filter is unstable; and the relationship between y_n and x_n is

$$y_n = x_n - 0.5x_{n-2} + 0.5x_{n-3} - 1.1y_{n-1} - 0.81y_{n-2} - 0.891y_{n-3}$$

FIR Filter

Introduction

- FIR Filters are filters with a finite impulse response.
- Assuming the impulse response of a filter is

$$[h_0, h_1, h_2, \cdots, h_{k-1}]$$

If the input sampled sequence to the filter is

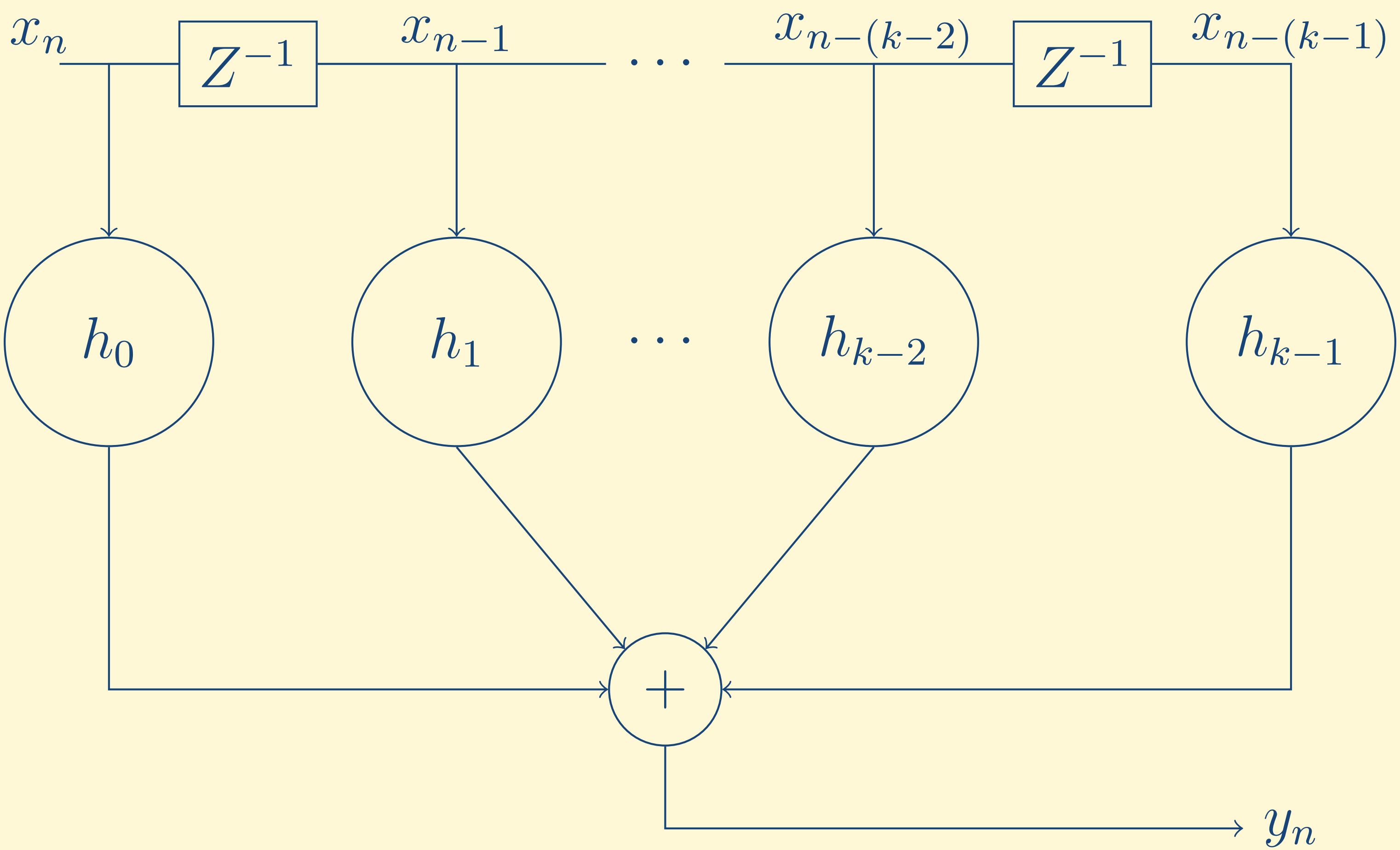
$$[x_n, x_{n-1}, x_{n-2} \cdots]$$

The output sequence is

$$y_n = x_n h_0 + x_{n-1} h_1 + \cdots + x_{n-k-1} h_{k-1}$$

The filter can be represented using a delay line as follows

FIR Notation



FIR Notation

- Remember, the sequence $h_0, h_1 \cdots h_{k-1}$ is just a sequence of k numbers stored in memory.
- The filter uses only the k most recent samples of the input data to compute the output.
  - Thus only the sequence $x_n, x_{n-1} \cdots , x_{n-(k-1)}$ is used (where x_n is the current sample).
  - Older samples are discarded.

Introduction

- The magnitude response of a filter is the magnitude of the transfer function on the unit circle on the $\mathbb{Z}$ Plane.
- It relates the magnitude of every normalised frequency $\frac{f}{F_s}$ at the output of the filter to the magnitude at the input.

Introduction

- The phase response of a filter is the phase of the transfer function on the unit circle on the $\mathbb{Z}$ Plane.
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- there are a maximum of 2^b possible numbers that can be represented

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Only positive numbers can be represented in this manner. The manner in which negative numbers are represented leads to different forms of fixed point arithmetic.

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- The main advantage of 2’s complement is that most operations are not dependent on the sign of the operands and furthermore are identical to operations on unsigned binary integers[1].
- An additional advantage is that if the final sum is within the range; even if partial sums result in overflows the final sum will be correct.

Examples are shown in the next two slides

[1] See http://en.wikipedia.org/wiki/Computer_numbering_formats

Binary Number	Value	Sign-Magnitude	1's Complement	2's Complement
0.000	0.000000	+0.000000	+0.000000	+0.000000
0.001	0.125000	+0.125000	+0.125000	+0.125000
0.010	0.250000	+0.250000	+0.250000	+0.250000
0.011	0.375000	+0.375000	+0.375000	+0.375000
0.100	0.500000	+0.500000	+0.500000	+0.500000
0.101	0.625000	+0.625000	+0.625000	+0.625000
0.110	0.750000	+0.750000	+0.750000	+0.750000
0.111	0.875000	+0.875000	+0.875000	+0.875000
1.000	1.000000	-0.000000	-0.875000	-1.000000
1.001	1.125000	-0.125000	-0.750000	-0.875000
1.010	1.250000	-0.250000	-0.625000	-0.750000
1.011	1.375000	-0.375000	-0.500000	-0.625000
1.100	1.500000	-0.500000	-0.375000	-0.500000
1.101	1.625000	-0.625000	-0.250000	-0.375000
1.110	1.750000	-0.750000	-0.125000	-0.250000
1.111	1.875000	-0.875000	-0.000000	-0.125000

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0011.	3.000000	+3.000000	+3.000000	+3.000000
0100.	4.000000	+4.000000	+4.000000	+4.000000
0101.	5.000000	+5.000000	+5.000000	+5.000000
0110.	6.000000	+6.000000	+6.000000	+6.000000
0111.	7.000000	+7.000000	+7.000000	+7.000000
1000.	8.000000	-0.000000	-7.000000	-8.000000
1001.	9.000000	-1.000000	-6.000000	-7.000000
1010.	10.000000	-2.000000	-5.000000	-6.000000
1011.	11.000000	-3.000000	-4.000000	-5.000000
1100.	12.000000	-4.000000	-3.000000	-4.000000
1101.	13.000000	-5.000000	-2.000000	-3.000000
1110.	14.000000	-6.000000	-1.000000	-2.000000
1111.	15.000000	-7.000000	-0.000000	-1.000000

Floating Point Numbers

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  ; and keeping the exponent as an unsigned binary integer.
- The number of bits b must be split into two; some bits for the mantissa and some for the exponent.

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Floating point representation results in rounding errors for both addition and multiplication whereas in fixed point notation, only multiplication results in rounding errors.

[2] See the IEEE 754 standard for 32 bit floating point numbers

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The addition and multiplication can be done using the instructions of the microprocessor or using new routines (written by the programmer/engineer).

Digital Filters

Finite Precision Effects

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- The addition and multiplication of finite precision numbers will also lead to rounding of the results.
- Also IIR filters (filters with feedback) are more sensitive. The feedback of rounding errors can cause oscillations in computation. **This effect is known as Limit Cycle**.

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Thus a strategy for implementing filters is to

- Cascade many smaller filters which have poles (and zeros) as far apart as possible.
- Have in parallel many smaller filters with the same criteria above.

In practice the cascade form is used as it is more robust to quantisation effects. Also poles closer to the origin are implemented first.
Typically filters with two or fewer poles are implemented.

Strategy

If the transfer function has m poles and zeros, with zeros at locations α and poles at locations β; the transfer function can be written as

$$H(Z) = \frac{(Z-\alpha_0)(Z-\alpha_1)\cdots(Z-\alpha_{m-1})}{(Z-\beta_0)(Z-\beta_1)\cdots(Z-\beta_{m-1})}$$

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or the parallel form

$$H(Z) = \sum \frac{c_{k0} + c_{k1}Z^{-1}}{1 + a_{k1}Z^{-1} + a_{k2}Z^{-2}}$$

Strategy

There are many software tools that can be used to select the best strategy. In practice, MATLAB® is normally used. **Remember there is no optimum method that works for all types of filters.**

Questions

Filters and the $\mathbb{Z}$ Transform

Determine the $\mathbb{Z}$ Transform of the following digital filters; and determine the transfer function of the filters in the frequency domain.

1. $y_n = x_n + 2x_{n-1} + 2x_{n-3} + x_{n-4}$
2. $y_n + y_{n-1} = x_n + 2x_{n-1} + 2x_{n-3} + x_{n-4}$

Sampling and $\mathbb{Z}$ Transform

Suppose a digital system has at its input (before sampling)

$$25\cos(2\pi 10 t)$$

What angle θ in the Z domain would the signal above occur at if the signal is sampled at

1. 100 Hz
2. 50 Hz
3. 25 Hz
4. 12 Hz

Given two Digital Filters with transfer functions

$$\frac{Y}{X} = Z^2 + 2Z + 3$$

$$\frac{Y}{X} = Z^{-2} + 2Z^{-1} + 3$$

Determine the *difference equation*[3]. Sketch the structures of these filters.

[3] The Difference Equation is the relationship between y_n and x_n

Normalised Filters

Given a digital filter with transfer function

$$1 + 2Z^{-1} + 3Z^{-2} + 2Z^{-3} + Z^{-4}$$

Determine the magnitude and phase response of the filter at a normalised frequency[4] of 0.25

4 $\frac{f}{Fs}$

A Digital Filter has transfer function

$$\frac{Y}{X} = Z^{-1} - 1$$

Given this is used to filter a signal

$$25\cos(2\pi 10t)$$

which is sampled at 50 Hz; **calculate**

1. The value of Z on the unit circle that represents this frequency.
2. The magnitude of the transfer function at this particular Z.
3. The phase of the transfer function at this particular value of Z.
4. Determine the group delay (in number of samples) at this Z.